SCMS SCHOOL OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CSD415 PROJECT PHASE I

AI Driven IoT Framework for Energy Optimization in Ductless HVAC Systems

PROBLEM STATEMENT

To develop a smart AI-powered IoT system that optimizes energy consumption in ductless air conditioners for small offices and residences.

SPECIFIC OBJECTIVES

1. To design an IoT system that monitors in real time the environmental and HVAC system parameters (temperature, voltage, current).
2. To implement Nearest Neighbor-based Density-Based Spatial Clustering of Applications with Noise algorithm (NN-DBSCAN) for optimization.
3. To create a real time dashboard for data visualization and updates on information and decisions .

ABSTRACT

This project proposes an IoT-based smart control system for ductless HVAC units using AI-driven analytics. By combining real-time sensor data with NN-DBSCAN clustering, the system detects anomalies and enables adaptive control and energy optimization. The Dashboard Interface provides users with real-time monitoring and control capabilities, ensuring seamless

interaction with the system. Unlike existing models that rely on complex machine learning setups or expensive cloud infrastructure, this project delivers a lightweight and modular design ideal for homes and small offices. It outperforms other models by balancing practical AI with real-world application. The proposed IoT-based smart control system for ductless HVAC units integrates several key components to enhance energy efficiency and user experience.

TEAM MEMBERS

1. ATHUL KRISHNA S ROLL NO : SCS/9938/22
2. LUKE ROMMEL MYLAADY ROLL NO : SCS/9935/22
3. JAMES RENIL ROLL NO : SCS/10118/22
4. JOHAN ITTY JONES ROLL N0 : SCS/9894/22

SIGNATURES

SIGNATURE OF PROJECT GUIDE:

SIGNATURE OF CO-ORDINATOR: